What goes where: equations, numbers, and section lines

Nothing from the engineer pages goes on the full lecture spreads. Each line below has a home in a reserved page. Say each line before you write it.

A · Equations & key numbers page (p3–4)

Box each formula. One worked number beside it. Units always.

$v = \sqrt{(\mu/r)}$ circular speed · LEO 400 km: 7.67 km/s · GPS: 3.87 · GEO: 3.07

$v_{esc} = \sqrt{(2\mu/r)} = \sqrt{2} \times v_{circ}$ · from LEO ≈ 11 km/s

$T = 2\pi\sqrt{(r^3/\mu)}$ period · GEO: T = 86,164 s → r = 42,164 km → alt 35,786

$v^2 = \mu(2/r - 1/a)$ vis-viva · GTO perigee: 10.15 km/s

$TOF = \pi\sqrt{(a^3/\mu)}$ Hohmann · LEO→GEO 5.3 h · Earth→Mars 259 d

$\Delta v_{plane} = 2v\cdot\sin(\Delta i/2)$ · 1° at LEO ≈ 134 m/s · 28.5°→0° ≈ 3.8 km/s

$\Delta v = Isp\cdot g_0\cdot\ln(m_0/m_f)$ rocket eqn · Isp 450 s → 4.4 km/s per e-fold

$\sin(\delta/2) = 1/(1 + r_p\cdot v_\infty^2/\mu)$ flyby bend · Jupiter 10 km/s @ 500,000 km → 92°

$P = \sigma\cdot\varepsilon\cdot T^4$ radiator · 1 m² @ 300 K, ε 0.9 ≈ 410 W

$drop = \frac{1}{2}gt^2$ · $curve = d^2/(2R)$ · ISS 1 s: 4.3 m = 4.3 m

μ_{Earth} = 398,600 km³/s² · R_{Earth} = 6,371 (6,378 eq.) km · g at 400 km = 8.7 m/s²

GTO burns: +2.42 then +1.47 km/s = 3.89 · Mars: 3.6 leave + ~2 capture

Earth 29.8 km/s · Jupiter 13 km/s around the Sun · sunlight 1,361 W/m²

synodic Earth–Mars 780 d (26 mo) · phase angle 44° · GEO station-keeping ~50 m/s/yr

B · Section map pages (1 page after each section)

One mechanism sentence per scene, under that scene's box. These are the lines you say out loud when someone asks "but how?"

S1 · falling & missing: drops 4.3 m/s, ground curves 4.3 m/s → never lands; engines only for pointing + drag

S1 · shelves: gravity ∝ $1/r^2$ → higher = slower; GEO is where T = 1 sidereal day; GPS = time delays → spheres, 4 sats

S1 · ladder: each rung = more Δv to slow down; flyby needs no capture burn

S2 · vacuum: only exit for heat is radiation ∝ T^4 → big hot panels; outgas → cold optics; sunlit + vs shaded − → arcs

S2 · air: stirred mix to 100 km, then sorted by weight: N_2/O_2 → O → He → H; UV splits O_2 → hungry O at 200–600

S3 · hoop: prograde @ perigee raises apogee (a↑ e↑); sideways @ node tilts i (expensive); radial rotates ω

S3 · orbits: LEO → GTO (+2.42) → apogee kick (+1.47) → GEO

S3 · fuel bill: vis-viva → Δv → rocket eqn → fuel grows exponentially → 20-day Mars impossible

S3 · free lunch: planet frame: same speed, bent δ; Sun frame: + planet's 13 km/s → faster; planet pays 10^{-24} m/s

C · Word bank (back pages) — 15 new lines

Same shape as the rest: p__ · term = also called · short meaning

centripetal force · gravity = the string

station-keeping · small burns, holds the slot

reboost · ISS +2 km/month

free fall · why you float

graveyard orbit · GEO + 300 km, retired

state vector · x y z vx vy vz at an epoch

TLE · public element format

Oberth effect · burn where fastest

Isp · engine efficiency, seconds

porkchop plot · Δv vs dates map

Stefan–Boltzmann · P ∝ T^4

ε / α · radiates well / absorbs little

heat pipe · boil-condense loop, no pump

photoelectric effect · UV knocks electrons off

GTO burns · 2.42 + 1.47 km/s

Question bank cards: the “but how?” questions

Back pages, Q on the left page, A on the facing right. These are the ones a friend who knows a little would ask. 20 cards.

Q (left page)

1 · How does a satellite stay up with its engines off?
2 · ISS drops 4.3 m every second. Why isn't it lower?
3 · What does the rocket actually provide: height or speed?
4 · Why do astronauts float if gravity is 90%?
5 · Why is a higher orbit slower?
6 · Where does 35,786 km come from?
7 · Two burns from LEO to GEO: where, which way, how big?
8 · How does GPS turn a radio signal into a position? Why 4?
9 · Why do GEO launches happen near the equator?
10 · Burn prograde at perigee: which elements change, which don't?
11 · Why is a plane change so expensive? Where do you do it?
12 · Where do published orbital elements come from?
13 · How does a radiator get rid of heat with no air?
14 · Why is one side of a spacecraft +120 °C and the other −150 °C?
15 · How does charge build up and what stops the arc?
16 · Walk the LEO→GEO Hohmann numbers.
17 · Why can't we reach Mars in 20 days? (use the rocket equation)
18 · How is Lambert's problem actually solved?
19 · In Jupiter's frame what changes in a flyby? In the Sun's?
20 · What three things set how much a flyby bends you?

A (facing right page)

1 · It's moving sideways at 7.7 km/s; it falls and the ground curves away equally. Gravity is the centripetal force.
2 · In that second it moved 7.7 km and Earth curved away 4.3 m. Drop = curve.
3 · Speed. 7.7 km/s sideways. Height is cheap.
4 · Free fall: nothing pushes back on you. Weight is the push, not the pull.
5 · Gravity ∝ $1/r^2$: less pull → less speed needed, and a bigger circle. T ∝ r^1.5.
6 · Solve T = 86,164 s in $T = 2\pi\sqrt{(r^3/\mu)}$: r = 42,164 km, minus 6,378 = 35,786.
7 · Prograde at perigee +2.42 km/s (raises apogee), prograde at apogee +1.47 (circularizes). 5.3 h apart.
8 · Delay × c = distance → sphere; 3 spheres = a point; 4th fixes the phone's clock.
9 · Launch latitude = orbit tilt; GEO needs 0°; plane change costs $2v\cdot\sin(\Delta i/2)$, ~3.8 km/s from Florida.
10 · a↑ and e↑ (apogee rises). i, Ω, ω unchanged.
11 · $\Delta v = 2v\cdot\sin(\Delta i/2)$: 1° ≈ 134 m/s at LEO. Do it at apogee where v is small, or launch into the right plane.
12 · Tracking gives position + velocity (state vector) at an epoch; geometry converts to a e i Ω ω ν; published as a TLE.
13 · Radiation: $P = \sigma\varepsilon T^4$. Big, hot, high-ε panels; heat pipes carry heat to them.
14 · No convection to even it out: sunlit side settles where absorbed = radiated (~120 °C); shade radiates to 3 K space.
15 · UV kicks electrons off sunlit surfaces (+); shade collects plasma electrons (−); kV gap arcs. Fix: conductive coatings + bonding straps.
16 · 7.73 → 10.15 at perigee (+2.42); 1.61 at apogee → 3.07 (+1.47); a = 24,418 km; TOF 5.3 h.
17 · ~60 km/s needed; e^(60/4.4) ≈ 800,000 → 99.9999% propellant. 3.6 km/s (Hohmann) needs 56%.
18 · Infinitely many ellipses fit 2 points; TOF depends on a; iterate a until TOF matches; velocities fall out.
19 · Jupiter frame: speed same, direction bent by δ. Sun frame: add Jupiter's 13 km/s → faster (behind) or slower (in front).
20 · Approach speed v_∞, closest approach r_p, planet's μ: $\sin(\delta/2) = 1/(1 + r_p v_\infty^2/\mu)$.